

Jiyeon Han

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Research Interests

Human-Computer Interaction · Accessibility · Human-Centered AI · Human-AI Interaction

Education

Ewha Womans University

Seoul, Korea

Master of Science (Computer Science and Engineering)

2022-2024

Thesis: MuseForge: AI-Based Audio Generation Model System for Video Creation

Advisor: Uran Oh

Ewha Womans University

Seoul, Korea

Bachelor of Science in Engineering (Computer Science & Engineering)

2016-2022

Capstone Project: Music Assembler, a touch-sensor musical instrument

Research Experience

Human Computer Interaction Lab, Ewha Womans University

Researcher (Advisor: Uran Oh)

2022–2024; 2025–Present

- **Accessible AI Photography (ongoing):** Developed an iOS prototype and conducted user research on AI-assisted photography for blind and low-vision users; analyzed interviews and interaction logs to inform accessible interaction design.
- **ForceKeyboard:** Designed a 26-participant study of force-sensitive, eyes-free typing while sitting and walking; analyzed typing speed, accuracy, and interview feedback.
- **AscleAI:** Designed and conducted interviews and a design probe with six clinicians; analyzed feedback and implemented RAG for clinical notes with LangChain and LlamaIndex.
- **MuseForge:** Independently compared an AI music generator and a music search engine (n=16); analyzed task times, prompts, and interviews. Identified difficulties in prompt formulation and built an LLM-assisted prompting prototype to address them.
- **TogGrid:** Built a low-cost tactile toggle-button interface; designed a shape-reproduction study with 12 blind and low-vision participants; analyzed time, accuracy, and interviews.

Publications

Peer-Reviewed Journal & Conference Papers

- Cavazos Quero, L., **Han, J.**, & Oh, U. (2026). ForceKeyboard: Eyes-Free Text Entry With a Force Sensitive Soft Keyboard for Wearable Devices. IEEE Transactions on Haptics, 19(2), 390–401. [[paper](#)]
- **Han, J.**, Kim, Y., Joh, H., Lee, J., & Oh, U. (2023). MoodChartBot: Design and Implementation of a Mood Chart Application Using AI Chatbot. Journal of Korea Multimedia Society, 26(3), 503–508. [[paper](#)]
- Kim, Y.*, **Han, J.***, & Oh, U. (2023). Exploring Low-Cost Grid-Based Tactile Instruments for Understanding and Reproducing Shapes for People with Visual Impairments. International Journal of Advanced Smart Convergence, 12(3), 127–140. [[paper](#)]

Publications (continued)

- **Han, J.**, Moon, S.J., Park, D.E., & Oh, U. (2019). Music Assembler: Integrated Virtual Musical Instrument with Block-Typed Touch Sensor. In Proceedings of the Korea Software Congress, 1687–1689. [[paper](#)]

Peer-Reviewed Extended Abstracts & Posters

- **Han, J.***, Park, J.*, Huh, J., Oh, U., Do, J., & Kim, D. (2024). AscleAI: A LLM-based Clinical Note Management System for Enhancing Clinician Productivity. In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24). ACM. [[paper](#)]
- **Han, J.**, Yang, E., & Oh, U. (2024). Understanding the Use of AI-Based Audio Generation Models by End-Users. In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24). ACM. [[paper](#)]

* Equal contribution.

Research Methods & Technical Skills

Programming: Python, Swift, Java, JavaScript, C, SQL
Methods: Interviews, formative studies, usability testing, think-aloud, surveys
Analysis: Qualitative coding, affinity mapping, interaction-log and task-performance analysis
Prototyping: iOS/SwiftUI, Android, multimodal AI, LLM/RAG, Figma, Unity

Awards & Scholarships

HSpace HRR Scholarship	Feb 2026–Present
Boeing Scholarship Recipient	Aug 2024
Grand Prize, Ewha Graduate Student Convergence Research Forum	Feb 2023
Ewha Womans University Admissions Scholarship for Graduates	Mar 2022
Outstanding Capstone Design Project Award	Dec 2019

Teaching Experience

Graduate Teaching Assistant
C Programming; Artificial Intelligence (Spring 2023)
Automata Theory (Fall 2022); Linear Algebra (Spring 2022)
Supported labs, tutorials, projects, exam preparation, and grading.

Undergraduate Teaching Assistant, Linux Workshop
Summer 2019 · Assisted lab practice and grading.

Academic Service & Selected Activities

Reviewer, ACM CHI 2027, Papers track	2026
Student Volunteer, SIGGRAPH Asia 2025	2025
Member, Game Development Club KING	2017–2022
Collaborated on a student tycoon game released on Steam.	